

# GEORGE JOBI PERANGATTU

Tempe, AZ 85281 | (480) 742-9855 | gjobiper@asu.edu

## PROFESSIONAL SUMMARY

Robotics engineer pursuing an M.S. in Robotics and Autonomous Systems with hands-on experience debugging real robotic systems, analyzing sensor data, building diagnostics-oriented test sequences, and documenting experimental results. Background spans robot navigation, robot fleet reliability concepts, electromechanical system interaction, sensors, actuators, cameras, LiDAR, Python scripting, ROS/Gazebo simulation, controls, and computer vision. Strong fit for forward deployed robotics work involving warehouse-style operational environments, fleet triage, root cause analysis, failure analysis, on-site diagnostics, telemetry/logs review, data collection, and cross-functional handoffs.

## EDUCATION

M.S. Robotics and Autonomous Systems, AI Concentration, Arizona State University, Tempe, AZ - Graduating Jul 2026

B.S. Computer Science, Arizona State University, Tempe, AZ - Graduated May 2025

## TECHNICAL SKILLS

Robotics Diagnostics: Robot navigation, sensor fusion, cameras, LiDAR, RGB-D cameras, actuators, PID control, closed-loop control, kinematics, ROS/ROS 2, Gazebo, CARLA, URDF

Field and Data Analysis: Python, Bash fundamentals, log analysis, logs, sensor-data logging, data collection, experiment design, root cause debugging, failure analysis, technical documentation, issue reproduction, test procedures, process improvement

Software and Systems: C++, Linux/Ubuntu, Git, Docker, CMake, CI/CD, testing, debugging, GDB, algorithms, data structures

Perception and ML: PyTorch, computer vision, point cloud processing, object detection, segmentation, depth estimation, tracking, model evaluation

## ROBOTICS EXPERIENCE

Parrot MiniDrone: Autonomous Robot Navigation on Real Hardware, ASU Robotics Coursework - Spring 2026

- Built and tested autonomous navigation on a physical aerial robot, integrating inertial and visual sensor data with closed-loop PID control and diagnostic logs.
- Logged sensor data during automated test sequences and evaluated navigation performance to identify failure modes and reliability improvements.
- Engineered safety systems for real-world autonomous operation and validated behavior across simulation and hardware.

3D Motion Planning for Robot Navigation in Gazebo, ASU Robotics Coursework - Fall 2025

- Developed Python-based 3D RRT path planning and ROS/Gazebo validation for collision-free robot motion.
- Designed experiments across multiple environment configurations and documented motion-planning behavior and trajectory quality.

Efficient TransFuser: Multi-Task Sensor Fusion for Autonomous Navigation, ASU CSE 598 Research Project - Spring 2026

- Built a camera-LiDAR sensor fusion model for autonomous navigation, improving held-out loss by 8.7% while reducing model size by 70%.
- Developed data loading, training, and evaluation pipelines across navigation, depth estimation, segmentation, detection, and BEV mapping tasks.
- Analyzed FP16 GPU-training instability and debugged model behavior through reproducible experiments and organized metrics.

Spider CAD Robot: Multi-Legged Robot Design, ASU Robotics Coursework - Spring 2026

- Designed a multi-legged robot CAD model with constrained joints and prepared URDF export for simulation, controls, and mechanical-software integration work.

## WORK EXPERIENCE

Backend Developer Intern, DigiClips Media, Tempe, AZ - Aug 2024 to May 2025

- Developed Python/PyTorch computer vision models and optimized training/data pipelines for 10M+ daily queries, reducing latency by 45%.
- Improved detection accuracy by 30% through evaluation, debugging, and reusable code.

Full Stack Developer Intern, Odoo, Dubai, UAE - May 2025 to Aug 2025

- Refactored Python backend systems and collaborated with engineering teams on production-quality code.

Full Stack Developer Intern, TicketDex, Tempe, AZ - Jan 2025 to May 2025

- Built backend features and tests, reducing latency by 30% through algorithmic optimization.